DHCP Installation

First, install the Kea DHCP service and then open the configuration file.

```shell
sudo install kea
sudo nano /etc/kea/kea-dhcp4.conf
```

Below is what the configuration file looks like:

```shell
{
    "Dhcp4": {
        #interfaces used by the server to listen on
        "interfaces-config": {
            "interfaces": ["eth0"]
        },
        "control-socket": {
            "socket-type": "unix",
            "socket-name": "/run/kea/kea4-ctrl-socket"
        },

        #default lease database on server
        "lease-database": {
            "type": "memfile",
            "lfc-interval": 3600
        },

        #leasetime in seconds with a min of 600 (10 min) and a max of 7200 (2
hours)
        "valid-lifetime": 600,
        "max-valid-lifetime": 7200,

        #range of Assignable IP addresses
        "subnet4": [{
            "id": 1,
            "subnet": "192.168.1.0/24",
            "pools": [{
                "pool": "192.168.1.150 - 192.168.1.200"
            }],

            #specify host reservations based on the MAC address
            "reservations": [
                {
                    "hw-address": "00:15:5D:0A:7E:06",
                    "ip-address": "192.168.1.100",
                    "hostname": "ca1.yalpmis.com"
                }
            ],
```

```
            #options for specifying gateway, DNS servers, and the domain
            "option-data": [{
                    "name": "routers",
                    "data": "192.168.1.1"
                },
                {
                    "name": "domain-name-servers",
                    "data": "192.168.1.135"
                },
                {
                    "name": "domain-name",
                    "data": "yalpmis.com"
                }
            ]
        }]
    }
}
```

Check the configuration to see if it succeeded successfully.

```
Shell
kea-shell --host 127.0.0.1 --port 8000 --auth-user kea-api --auth-password $(cat
/etc/kea/kea-api-password) --service dhcp4 config-reload
```

Check the logs

```
Shell
journalctl -u kea-dhcp4-server
```

Restart the service.

```
Shell
systemctl restart kea-dhcp4-server
```

[How to install and configure isc-kea - Ubuntu Server documentation]

DNS Server Configuration

Install Berkley Internet Name Domain (BIND) and tools. Then, edit the options file in the /etc/bind directory.

```shell
sudo apt install bind9 dnsutils
sudo nano /etc/bind/named.conf.options
```

Below is the options file on the named configuration:

```shell
options {
	directory "/var/cache/bind";

	allow-query { localhost; 192.168.1.0/24; };

	// Configure forwarders
	forwarders {
		8.8.8.8;
		1.1.1.1;
	};

	// Other options
	listen-on { 192.168.1.135; };
	dnssec-validation auto;
};
```

Edit the local configuration of the server. This is where the forward and reverse lookup zones will be specified.

```shell
sudo nano /etc/bind/named.conf.local
```

Local configuration file:

```shell
//
// Do any local configuration here
//

zone "yalpmis.com" {
	type master;
	file "/etc/bind/zones/yalpmis.com.db";
};

zone "1.168.192.in-addr.arpa" {
```

```
        type master;
        file "/etc/bind/zones/db.192.168.1";
};
```

Create a new directory called zones/ and change to it. Then, create a new file for the forward lookup zone.

```Shell
mkdir -p zones
cd zones/
sudo nano db.yalpmis.com
```

Example configuration of the forward zone file:

```Shell
;
; BIND data file for yalpmis.com
;
$TTL    604800
@       IN      SOA     ns1.yalpmis.com. admin.yalpmis.com. (
                              2             ; Serial
                         604800             ; Refresh
                          86400             ; Retry
                        2419200             ; Expire
                         604800 )     ; Negative Cache TTL
;
@       IN      NS      ns1.yalpmis.com.
@       IN      A       192.168.1.135
ns1     IN      A       192.168.1.135
```

Now, create the reverse lookup zone file.

```Shell
sudo nano db.192.168.1
```

Reverse lookup zone file:

```Shell
$TTL    604800
@       IN      SOA     ns1.yalpmis.com. admin.yalpmis.com. (
                              2             ; Serial
                         604800             ; Refresh
                          86400             ; Retry
```

```
                    2419200            ; Expire
                     604800 )    ; Negative Cache TTL
;
@       IN      NS      ns1.yalpmis.com.
135     IN      PTR     ns1.yalpmis.com.
```

Check the configuration file and then restart the service.

```shell
sudo named-checkconf
sudo named-checkzone yalpmis.com /etc/bind/zones/db.yalpmis.com
sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/zones/db.192.168.1
sudo systemctl restart named
```

[Setting Up a DNS Server on Ubuntu: A Guide for Network Engineer | by Syed Asif | Medium]

DNSSEC

Create keys Zone Signing Key (ZSK) and Key Signing Key (KSK):

```
Shell
cd /var/lib/bind --> directory to store keys
sudo dnssec-keygen -a RSASHA256 -b 2048 -n ZONE yalpmis.com --> pub (.key) and
priv (.private) key pair for ZSK
sudo dnssec-keygen -f KSK -a RSASHA256 -b 2048 -n ZONE yalpmis.com --> pub and
priv keys for KSK
```

Add to the zone file (/var/lib/bind/zones/yalpmis.com.db):

```
Shell
;
; BIND data file for yalpmis.com
;
$TTL   604800
@      IN     SOA    yalpmis.com. admin.yalpmis.com. (
                           2             ; Serial
                      604800             ; Refresh
                       86400             ; Retry
                     2419200             ; Expire
                      604800 )     ; Negative Cache TTL
;
@      IN     NS     yalpmis.com.
@      IN     A      192.168.1.135
www    IN     A      192.168.1.135
test1  IN     CNAME  server
;
$INCLUDE "Kexample.com.+008+12345.key"
$INCLUDE "Kexample.com.+008+67890.key"
```

Sign the zone file:

```
Shell
dnssec-signzone -A -3 $(head -c 1000 /dev/urandom | sha1sum | cut -b 1-16) \ -N
increment -o yalpmis.com -t db.yalpmis.com
```

Add to local configuration (named.conf.local):

```
Shell
zone "example.com" {
       type master;
       file "/var/lib/bind/db.example.com.signed";
       inline-signing yes;
};
```

Restart the service:

```shell
sudo systemctl restart named
```